

Install Splunk Universal Forwarder in Linux EC2 Instance

Summary

This document explains the steps involved in installing Splunk Universal Forwarder on a Linux Server.

1. Connect to the Linux EC2 instance using SSH Client Command.

```
ssh -i "datasource2.pem" ec2-user@ec2-<ip-address>-<location>.compute.amazonaws.com
```

```
anusharc@MacBookPro ~ % cd Downloads/keypairs
anusharc@MacBookPro keypairs % ssh -i "datasource2.pem" ec2-user@ec2-54-252-237-14.ap-southeast-2.compute.amazonaws.com
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Tue Apr  7 00:36:41 2026 from 43.245.153.87
[ec2-user@ip-172-31-7-24 ~]$
```

2. Set Hostname and Reboot the system (Optional)

- 1) Become a Root user

```
sudo su
```

- 2) Change hostname(Optional) and Reboot to check the changes

```
hostnamectl set-hostname <Component_Name>
reboot
```

```
[ec2-user@ip-172-31-7-24 ~]$ sudo su
[root@ip-172-31-7-24 ec2-user]# hostnamectl set-hostname LinuxServer
[root@ip-172-31-7-24 ec2-user]# reboot
[root@ip-172-31-7-24 ec2-user]# Connection to ec2-54-252-237-14.ap-southeast-2.compute.amazonaws.com closed by remote host.
Connection to ec2-54-252-237-14.ap-southeast-2.compute.amazonaws.com closed.
anusharc@MacBookPro keypairs %
```

3. Create new user (splunk), directories & permissions

- 1) Create user & group

```
useradd splunk
groupadd splunk
```

- 2) Create directory

```
mkdir /opt/splunkforwarder
```

- 3) Change the ownership of the created directory to the new user "splunk"

```
chown -R splunk:splunk /opt/splunkforwarder
```

```
[ec2-user@LinuxServer ~]$ sudo su
[root@LinuxServer ec2-user]# useradd splunk
[root@LinuxServer ec2-user]# groupadd splunk
groupadd: group 'splunk' already exists
[root@LinuxServer ec2-user]# mkdir /opt/splunkforwarder
[root@LinuxServer ec2-user]# chown -R splunk:splunk /opt/splunkforwarder/
```

- 4) Verify the ownership of the directory

```
ll /opt/
```

```
[root@LinuxServer ec2-user]# ll /opt/
total 0
drwxr-xr-x. 2 splunk splunk 6 Apr  7 00:46 splunkforwarder
[root@LinuxServer ec2-user]#
```

4. Install wget

```
yum install wget
```

```
[root@LinuxServer ec2-user]# yum install wget
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.
Red Hat Enterprise Linux 9 for x86_64 - AppStream from RHUI (RPMs)        67 MB/s | 86 MB    00:01
Red Hat Enterprise Linux 9 for x86_64 - BaseOS from RHUI (RPMs)       56 MB/s | 110 MB   00:01
Red Hat Enterprise Linux 9 Client Configuration                       43 kB/s | 4.1 kB   00:00
Dependencies resolved.
=====
Package                Architecture      Version           Repository          Size
=====
Installing:
wget                   x86_64            1.21.1-8.el9_4    rhel-9-appstream-rhui-rpms  789 k
Transaction Summary
=====
Install 1 Package

Total download size: 789 k
Installed size: 3.1 M
Is this ok [y/N]: y
Downloading Packages:
wget-1.21.1-8.el9_4.x86_64.rpm                                          3.9 MB/s | 789 kB    00:00
-----
Total                                                                    3.4 MB/s | 789 kB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                :
  Installing                : wget-1.21.1-8.el9_4.x86_64                1/1
  Running scriptlet: wget-1.21.1-8.el9_4.x86_64                        1/1
  Verifying                 : wget-1.21.1-8.el9_4.x86_64                1/1
Installed products updated.

Installed:
  wget-1.21.1-8.el9_4.x86_64

Complete!
[root@LinuxServer ec2-user]#
```

5. Splunk Universal Forwarder installation

- 1) Switch to splunk user

```
sudo su - splunk
```

- 2) Navigate to splunk user home directory

```
cd /home/splunk
```

- 3) Download Splunk Universal Forwarder based on the version you want.

You can get this command from:

https://www.splunk.com/en_us/download/universal-forwarder.html?locale=en_us

The screenshot displays the Splunk Universal Forwarder download page. It lists various download options for different architectures and operating systems. A tooltip is visible over the 'Copy wget link' button for the 9.4.9 ARM 64-bit version, showing the command: `wget -O splunkforwarder-9.4.9-03bb451d4e07-linux-amd64.tgz 'https://download.splunk.com/products/universalforwarder/releases/9.4.9/linux/splunkforwarder-9.4.9-03bb451d4e07-linux-amd64.tgz'`

```
wget -O splunkforwarder-9.4.9-xxxx-linux-amd64.tgz
"https://download.splunk.com/products/universalforwarder/re
leases/9.4.9/linux/splunkforwarder-9.4.9-xxxx-linux-
amd64.tgz"
```

```
[root@LinuxServer ec2-user]# sudo su - splunk
[splunk@LinuxServer ~]$ cd /home/splunk
[splunk@LinuxServer ~]$ wget -O splunkforwarder-9.4.9-03bb451d4e07-linux-amd64.tgz "https://download.splunk.com/prod
ucts/universalforwarder/releases/9.4.9/linux/splunkforwarder-9.4.9-03bb451d4e07-linux-amd64.tgz"
--2026-04-07 01:02:32-- https://download.splunk.com/products/universalforwarder/releases/9.4.9/linux/splunkforwarde
r-9.4.9-03bb451d4e07-linux-amd64.tgz
Resolving download.splunk.com (download.splunk.com)... 3.175.115.82, 3.175.115.8, 3.175.115.18, ...
Connecting to download.splunk.com (download.splunk.com)|3.175.115.82|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 68853062 (66M) [binary/octet-stream]
Saving to: 'splunkforwarder-9.4.9-03bb451d4e07-linux-amd64.tgz'

splunkforwarder-9.4.9-03bb45 100%[=====] 65.66M 19.2MB/s in 3.4s
2026-04-07 01:02:36 (19.2 MB/s) - 'splunkforwarder-9.4.9-03bb451d4e07-linux-amd64.tgz' saved [68853062/68853062]

[splunk@LinuxServer ~]$
```

4) Extract the tar package

```
tar -xvf splunkforwarder-9.4.9-03bb451d4e07-linux-amd64.tgz  
-C /opt/
```

```
splunkforwarder/etc/system/default/web-features.conf  
splunkforwarder/etc/system/default/server.conf  
splunkforwarder/etc/system/default/metric_alerts.conf  
splunkforwarder/etc/system/default/authentication.conf  
splunkforwarder/etc/system/default/app.conf  
splunkforwarder/etc/system/default/alert_actions.conf  
splunkforwarder/etc/system/default/messages.conf  
splunkforwarder/etc/system/default/authorize.conf  
splunkforwarder/etc/system/default/source-classifier.conf  
splunkforwarder/etc/system/default/field_filters.conf  
splunkforwarder/etc/system/default/outputs.conf  
splunkforwarder/etc/system/default/sourcetypes.conf  
splunkforwarder/etc/system/default/limits.conf  
splunkforwarder/etc/system/default/telemetry.conf  
splunkforwarder/etc/system/default/audit.conf  
splunkforwarder/etc/system/default/ipc_broker.conf  
splunkforwarder/etc/system/default/visualizations.conf  
splunkforwarder/etc/system/default/conf.conf  
splunkforwarder/etc/system/default/restmap.conf  
splunkforwarder/etc/system/default/literals.conf  
splunkforwarder/etc/system/default/inputs.conf  
splunkforwarder/etc/system/default/default-mode.conf  
splunkforwarder/etc/system/default/agent_management.conf  
splunkforwarder/etc/system/default/federated.conf  
splunkforwarder/etc/system/default/web.conf  
splunkforwarder/etc/system/default/procmon-filters.conf  
splunkforwarder/etc/system/default/props.conf  
splunkforwarder/etc/system/default/livetail.conf  
splunkforwarder/etc/system/default/transforms.conf  
splunkforwarder/etc/system/default/global-banner.conf  
splunkforwarder/etc/system/default/health.conf  
splunkforwarder/etc/system/default/metric_rollups.conf  
splunkforwarder/etc/system/local/  
splunkforwarder/etc/system/local/README  
splunkforwarder/etc/disabled-apps/  
splunkforwarder/etc/disabled-apps/README  
splunkforwarder/etc/deployment-apps/  
splunkforwarder/etc/deployment-apps/README  
splunkforwarder/etc/manager-apps/  
splunkforwarder/etc/manager-apps/_cluster/  
splunkforwarder/etc/manager-apps/_cluster/default/  
splunkforwarder/etc/manager-apps/_cluster/default/indexes.conf  
splunkforwarder/etc/manager-apps/_cluster/local/  
splunkforwarder/etc/manager-apps/_cluster/local/README  
[splunk@LinuxServer ~]$
```

5) Navigate to splunk bin directory

```
cd /opt/splunkforwarder/bin
```

6) Start the Splunk with accepting license & set the default username and password

```
./splunk start -accept-license
```

Note: In above step, installer will ask you to create username and password, please keep these credentials safe (These are the admin credentials which you will use to Manage Splunk).

```
[splunk@LinuxServer ~]$ cd /opt/splunkforwarder/bin
[splunk@LinuxServer bin]$ ./splunk start --accept-license
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunk:splunk /opt/splunkforwarder"

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: adminuf
Password must contain at least:
    * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Creating unit file...
Current splunk is running as non root, which cannot operate systemd unit files.
Please create it manually by 'sudo splunk enable boot-start' later.
Failed to create the unit file. Please do it manually later.

Splunk> The IT Search Engine.

Checking prerequisites...
  Checking mgmt port [8089]: open
    Creating: /opt/splunkforwarder/var/lib/splunk
    Creating: /opt/splunkforwarder/var/run/splunk
    Creating: /opt/splunkforwarder/var/run/splunk/appserver/i18n
    Creating: /opt/splunkforwarder/var/run/splunk/appserver/modules/static/css
    Creating: /opt/splunkforwarder/var/run/splunk/upload
    Creating: /opt/splunkforwarder/var/run/splunk/search_telemetry
    Creating: /opt/splunkforwarder/var/run/splunk/search_log
    Creating: /opt/splunkforwarder/var/spool/splunk
    Creating: /opt/splunkforwarder/var/spool/dirmoncache
    Creating: /opt/splunkforwarder/var/lib/splunk/authDb
    Creating: /opt/splunkforwarder/var/lib/splunk/hashDb
    Creating: /opt/splunkforwarder/var/run/splunk/collect
    Creating: /opt/splunkforwarder/var/run/splunk/sessions
New certs have been generated in '/opt/splunkforwarder/etc/auth'.
  Checking conf files for problems...
    Done
  Checking default conf files for edits...
  Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-9.4.9-03bb451d4e07-linu
x-amd64-manifest'
    All installed files intact.
    Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
PYTHONHTTPSVERIFY is set to 0 in splunk-launch.conf disabling certificate validation for the httplib and urllib lib
aries shipped with the embedded Python interpreter; must be set to "1" for increased security
Done

[splunk@LinuxServer bin]$ █
```

6. Configure Splunk to run at boot time

- 1) Exit from splunk user and sign in as root user

```
exit
sudo su
```

```
[splunk@LinuxServer bin]$ exit
logout
[root@LinuxServer ec2-user]# sudo su
[root@LinuxServer ec2-user]# █
```

- 2) Enable boot start

```
/opt/splunkforwarder/bin/splunk stop
/opt/splunkforwarder/bin/splunk enable boot-start -user
splunk
/opt/splunkforwarder/bin/splunk start
```

```
[root@LinuxServer ec2-user]# /opt/splunkforwarder/bin/splunk stop
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunk:splunk /opt/splunkforwarder"
Stopping splunkd...
Shutting down. Please wait, as this may take a few minutes.
.
Stopping splunk helpers...

Done.
[root@LinuxServer ec2-user]# /opt/splunkforwarder/bin/splunk enable boot-start -user splunk
Systemd unit file installed by user at /etc/systemd/system/SplunkForwarder.service.
Configured as systemd managed service.
[root@LinuxServer ec2-user]# /opt/splunkforwarder/bin/splunk start
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunk:splunk /opt/splunkforwarder"

Splunk> The IT Search Engine.

Checking prerequisites...
  Checking mgmt port [8089]: open
  Checking conf files for problems...
  Done
  Checking default conf files for edits...
  Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-9.4.9-03bb451d4e07-Linu
x-amd64-manifest'
  All installed files intact.
  Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Done

[root@LinuxServer ec2-user]# █
```

The Splunk Universal Forwarder is successfully installed on Linux Server.

7. Next Steps

Connect Universal Forwarder with Heavy (Intermediate) Forwarder – Refer **07b_Connect_UF_with_HF_V1.pdf**.